Tirumala Teja

Data Scientist

940-977-0152 | yegineni111@gmail.com | LinkedIn | GitHub | Portfolio

Professional Summary

Data Scientist with nearly 4 years of experience developing machine learning, predictive analytics, and data-driven solutions for enterprise environments. Proficient in Python, SQL, Scikit-learn, XGBoost, PySpark, and Databricks, with hands-on experience building demand forecasting, customer segmentation, recommendation, and classification models. Experienced in developing automated ETL and feature-engineering pipelines, analyzing large-scale structured and unstructured datasets, and applying statistical modeling, hypothesis testing, and hyperparameter optimization. Skilled in end-to-end machine learning workflows, including model development, validation, experiment tracking, deployment, monitoring, and drift detection using MLflow and cloud platforms such as AWS, Azure, and GCP. And experienced at translating complex analytical findings into actionable business insights through Power BI dashboards and collaboration with engineering, product, merchandising, and supply-chain teams.

Education

University of North Texas — **Denton, TX**
Master of Science in Data Science

Relevant Coursework: Data Science, Machine Learning, Deep Learning, Operations Research, Optimization, NLP, Data Engineering, MLOps, and Artificial Intelligence

Certifications

IBM Data Science Professional Certificate

Microsoft Certified: Azure AI Engineer Associate

AWS Certified Solutions Architect –Associate

Coursera Machine Learning Specialization

Experience

EthicalZen.ai — **United States**
Data Scientist — *Dec 2024 – Present*

- Developed machine learning and NLP models to detect PII exposure, toxic content, prompt-injection attacks, and hallucinations in LLM prompts and responses, improving policy-violation detection accuracy by 22%.
- Built automated data ingestion, preprocessing, and feature-engineering pipelines using Python, SQL, PySpark, and Databricks to analyze more than 8 million AI interactions per month, reducing data preparation time by 60%.
- Designed risk-scoring models to classify AI requests by compliance severity and recommend enforcement actions, reducing manual review requirements by 35%.
- Developed evaluation frameworks for LLM safety and compliance using precision, recall, F1-score, false-positive rate, toxicity scores, and hallucination metrics across OpenAI, Claude, and other AI models.
- Implemented model validation, hyperparameter tuning, drift detection, and MLflow experiment tracking, reducing model-release cycles from three weeks to six days and improving reproducibility across development environments.
- Created Power BI dashboards for policy violations, PII incidents, prompt-injection attempts, model drift, and compliance KPIs, reducing audit-report preparation time by 45% and supporting SOC 2, HIPAA, and EU AI Act readiness.

Cyberspace Technologies — **India**
Data Scientist — *Jan 2022 – Dec 2023*

- Developed predictive models using Scikit-learn and XGBoost, applying advanced feature engineering and statistical techniques to improve forecasting accuracy by 22%.
- Applied machine learning and optimization techniques to agricultural forecasting problems, enabling data-driven planning and improved operational efficiency.
- Built scalable ETL pipelines for data ingestion, transformation, and feature extraction, supporting large-scale analytics and model-training workflows.
- Performed large-scale data analysis to identify trends, correlations, and business opportunities and translated findings into actionable recommendations.
- Designed interactive dashboards to communicate insights and help stakeholders make data-driven strategic decisions.
- Collaborated with cross-functional teams to define business problems and deliver scalable data science solutions.

Projects

AI-Powered Knowledge Retrieval System (RAG-Based)
Python, NLP, LLMs, FastAPI, Vector Search

- Designed a scalable Retrieval-Augmented Generation system integrating NLP, embeddings, and semantic search for intelligent knowledge extraction.
- Built end-to-end pipelines for document ingestion, indexing, and retrieval, reducing information-search time by approximately 50%.
- Integrated machine learning models and APIs to deliver real-time insights and support decision-making workflows.

Spotify Analytics Platform
PySpark, Databricks, AWS, Snowflake, Power BI, SQL

- Developed a large-scale PySpark pipeline on Databricks to process streaming datasets and perform high-performance analytics.
- Applied SQL-based analytics and data-modeling techniques to generate insights into user behavior and engagement trends.
- Designed dashboards to visualize trends and support business decision-making.

PM2.5 Air Quality Prediction System
Python, Machine Learning, Pandas, Scikit-learn, Data Analysis

- Built regression models using Scikit-learn to predict pollution levels, applying feature engineering and statistical analysis.
- Evaluated models using RMSE and R^2 metrics and optimized performance through hyperparameter tuning.

Technical Skills

Programming: Python, SQL, R , Bash
Machine Learning & AI: Scikit-learn, TensorFlow, PyTorch, XGBoost, Deep Learning, Model Evaluation, Feature Engineering
Advanced Analytics: Statistical Modeling, Hypothesis Testing, Optimization, Simulation, A/B Testing
Generative AI & NLP: NLP, Embeddings, Semantic Search, Retrieval-Augmented Generation, Prompt Engineering
Data Engineering: ETL Pipelines, PySpark, Databricks, Snowflake, Data Wrangling
Visualization: Power BI, pandas, numpy, Matplotlib, Seaborn, Exploratory Data Analysis(EDA)
Cloud Platforms: AWS, Azure, GCP
MLOps & Tools: Docker, Git, CI/CD, MLflow, Jupyter Notebook, VSCode, Claude Code, Cursor, GPT Codex

Key Strengths

- Algorithm Design and Scalable Model Development
- Large-Scale Data Processing and Cloud Computing
- Cross-Functional Collaboration and Communication
- Business Impact Through Data-Driven Insights
- Continuous Learning and Technical Mentorship